# **Apply to be a Science Corps Fellow** - Entry #3374

**Name**

Johannes Eckert

**Email**

johannes.eckert1048@t-online.de

**Gender**

Male

**Residence: City (State), Country**

Zurich, Switzerland

**Nationality**

German

**When did you or do you expect to complete your PhD?**

6th Nov 2025

**Please tell us briefly about your university education (institution, city, country, year)**

I began my university education at the University of Freiburg, Germany, where I completed my B.Sc. in Chemistry (2013–2016). My bachelor's thesis focused on rhodium-catalyzed organic transition metal catalysis. For my M.Sc. (2016–2019) in Biochemistry at Freiburg, I investigated the ABC transporter NosDFY in the nitrous oxide reductase maturation pathway. In 2020, I pursued an Erasmus exchange at Karolinska Institute in Stockholm, Sweden, studying the effects of diverse antibiotics on Salmonella biofilm growth. Since 2021, I have been pursuing a Ph.D. at the ETH Zürich, Switzerland, researching RiPP biosynthetic gene clusters, with a focus on lipopeptides, their natural diversity, and potential applications as pharmaceuticals and biocatalysts.

**Why would you like to be a Science Corps Fellow?**

From the beginning of my academic journey, I have been driven by curiosity and a desire to see science from multiple perspectives. Participating in conferences and meeting researchers from other countries, outside of Europe, opened my eyes to how scientific work can take place under diverse circumstances. These experiences inspired me to broaden my horizons, gain a more pragmatic view of science, and step outside the privileges I had often taken for granted.

My motivation is also deeply personal. I am the first in my family to attend university—growing up as the son of a goldsmith and an accountant, I had no role models in science or academia. What I did have was the encouragement of my parents to explore, stay curious, and pursue education. This support enabled me to study at a leading university and eventually complete a PhD. I hope to give back by offering the same encouragement to students who might otherwise not see a place for themselves in science.

I am particularly motivated by the opportunity to interact with students, ignite their curiosity, and share my

enthusiasm for the natural sciences. Encouraging critical thinking and fostering independent perspectives has always been exciting for me, as it not only stimulates learners but also challenges and enriches my own understanding.
Additionally, discussions with my wife, a teacher and educational researcher, have deepened my understanding of educational equity. Her passion for ensuring fair access to learning has influenced my own commitment to supporting students, particularly those who might otherwise lack opportunities to explore science.
Joining Science Corps represents for me the chance to combine international experience, teaching, and mentorship in a way that promotes curiosity, critical thinking, and access to science for all.

**In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.**

Beyond my academic training, I have gained experiences that have shaped my ability to connect with learners, adapt to different needs, and make learning engaging - skills I am eager to bring to the Science Corps Fellowship. Initially, I would not have envisioned myself in teaching or mentoring roles, but over time, I discovered how rewarding it can be.
My first experiences in education began as a youth group leader during my teenage years. I had no formal training, but I tried to lead by example and create a positive environment for children aged 9–12. We met weekly, played games, and organized trips, and years later, it still moves me when former group members fondly remember these experiences. It taught me early on the impact a supportive role model can have.
During my studies at the University of Freiburg, I worked as a teaching assistant for a chemistry practical course. Initially, I guided students directly in the lab, and later adapted the course to an online format during the COVID-19 lockdown. Despite the challenges, these experiences showed me how sharing knowledge and passion can truly help students engage with the material, even under challenging circumstances.
As a PhD student, I mentored several master's students and an exchange student, supporting them not only in practical laboratory work but also in developing critical thinking, independent planning, and creative problem-solving skills. It was gratifying to see how guiding students while allowing freedom sparked their own ideas, and how their growth into independent thinkers became a source of inspiration for me as well.
I have also benefited from regular conversations with my wife, a teacher and educational researcher, about pedagogy, classroom management, and curriculum design. These discussions have broadened my perspective on teaching and provided valuable insights into how to structure lessons and make learning engaging for diverse learners.
Together, these experiences have helped me develop adaptability, commitment to mentorship, and a deep enthusiasm for making science accessible and engaging.

**Where do you imagine yourself professionally in the future? Check all that apply.**

Scientist in academia
Doing science in industry
Administrative role in industry or academia

**Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.**

In the near term (~2 years), I would appreciate gaining industry experience to complement my academic background. I want to work in an environment with clear guidance, standard operating procedures, and structured workflows, where collaboration and shared responsibilities are central. Unlike the mostly solitary nature of academic projects, I look forward to contributing to a team working toward common goals and supporting other teams as needed. I am particularly interested in experiencing how scientific work can have a direct impact on human life and health by contributing to products that must meet governmental and regulatory guidelines. I also want to develop skills in process implementation to ensure compliance with these standards. In addition, I am eager to deepen my knowledge of computer science, data science, and related fields. I enjoy the challenge of creating and improving code, automating processes, reducing manual work, and streamlining workflows to make operations more efficient.

In the long term (~10 years), I envision myself in diverse roles that integrate my scientific, technical, and leadership skills. I could see myself in a university position, teaching and mentoring students while continuing research, or in a scientific team leadership role within a company, guiding projects and strategic decisions. Alternatively, I am interested in leveraging my expertise in computer science and data analysis to design workflows, automate processes, and extract actionable insights from complex datasets. I could also imagine combining these skills with business and management responsibilities, contributing to the strategic development of an organization. In all scenarios, my goal is to have a position that allows me to integrate scientific knowledge, technical skills, and leadership to make a meaningful impact on research, education, and organizational growth.

At the same time, I recognize that my professional path will also be shaped by my private life and the career of my wife. I remain open to opportunities that may arise along the way, as this openness has already guided me to my current PhD position. I am confident that staying receptive to such possibilities will continue to create meaningful and fulfilling directions for my career.

**Please upload your CV (.pdf)**

CV-Johannes-Eckert-2025.pdf

**How did you find out about Science Corps?**

An email list in your institution

Science Corps